

Untitled

Analytical Derivation of Temperature Profile for the 1D Linear Heat Equation

This section presents the full derivation of the temperature profile for the 1D linear heat equation introduced in Section X:

$$-\frac{d}{dx} \left(k(x) \frac{dT}{dx} \right) = q(x)$$

The domain is split into two regions:

- For $0 \leq x < 0.4$, the conductivity and source terms are $k_1(\mu)$ and $q_1(\beta)$.
- For $0.4 \leq x \leq 0.5$, the conductivity and source terms are $k_2(\mu)$ and $q_2(\beta)$.

Boundary conditions:

- $k_1(\mu) \frac{dT}{dx} \Big|_{x=0} = 0$
- $T(0.5) = 573.15, \text{K}$
- Continuity of temperature and heat flux at $x = 0.4$

Heat Equation in Each Region

Region 1 ($0 \leq x < 0.4$)

The governing equation is

$$-\frac{d}{dx} \left(k_1(\mu) \frac{dT_1}{dx} \right) = q_1(\beta)$$

With constant k_1 and q_1 , this becomes

$$k_1 \frac{d^2 T_1}{dx^2} = -q_1.$$

Integrate once:

$$k_1 \frac{dT_1}{dx} = -q_1 x + C_1.$$

Apply $k_1 \frac{dT_1}{dx} \Big|_{x=0} = 0$ to get $C_1 = 0$:

$$k_1 \frac{dT_1}{dx} = -q_1 x.$$

Integrate again:

$$T_1(x) = -\frac{q_1}{2 k_1} x^2 + C_2.$$

Region 2 ($0.4 \leq x \leq 0.5$)

The equation is

$$-\frac{d}{dx}\left(k_2(\mu)\frac{dT_2}{dx}\right) = q_2(\beta),$$

which simplifies to

$$k_2 \frac{d^2 T_2}{dx^2} = -q_2.$$

Integrate once:

$$k_2 \frac{dT_2}{dx} = -q_2 x + C_3.$$

Integrate again:

$$T_2(x) = -\frac{q_2}{2k_2} x^2 + \frac{C_3}{k_2} x + C_4.$$

Boundary and Continuity Conditions

Boundary Condition at $x = 0.5$

Given $T(0.5) = 573.15$ K,

$$T_2(0.5) = -\frac{q_2}{2k_2}(0.5)^2 + \frac{C_3}{k_2}(0.5) + C_4 = 573.15.$$

Continuity at $x = 0.4$

Temperature continuity:

$$T_1(0.4) = T_2(0.4)$$

$$-\frac{q_1}{2k_1}(0.4)^2 + C_2 = -\frac{q_2}{2k_2}(0.4)^2 + \frac{C_3}{k_2}(0.4) + C_4.$$

Heat-flux continuity:

$$k_1 \left. \frac{dT_1}{dx} \right|_{0.4} = k_2 \left. \frac{dT_2}{dx} \right|_{0.4}$$

$$k_1 \left(-\frac{q_1}{k_1}(0.4) \right) = k_2 \left(-\frac{q_2}{k_2}(0.4) + \frac{C_3}{k_2} \right)$$

which yields

$$C_3 = (q_2 - q_1) 0.4.$$

Solve for Constants

We have three equations:

1. **Boundary at 0.5:**

$$-\frac{q_2}{2k_2}(0.5)^2 + \frac{C_3}{k_2}(0.5) + C_4 = 573.15.$$

2. Temperature continuity at 0.4:

$$-\frac{q_1}{2k_1}(0.4)^2 + C_2 = -\frac{q_2}{2k_2}(0.4)^2 + \frac{C_3}{k_2}(0.4) + C_4.$$

3. Flux continuity:

$$C_3 = 0.4(q_2 - q_1).$$

Solving gives:

- $C_2 = \frac{0.08q_1}{k_1} + \frac{0.04q_1 + 0.005q_2}{k_2} + T_{0.5}$
- $C_3 = 0.4(q_2 - q_1)$
- $C_4 = \frac{0.2q_1 - 0.075q_2 + k_2T_{0.5}}{k_2}$

where $T_{0.5} = 573.15$ K.

Final Temperature Profile

- For $0 \leq x < 0.4$:

$$T_1(x) = -\frac{q_1}{2k_1}x^2 + C_2.$$

- For $0.4 \leq x \leq 0.5$:

$$T_2(x) = -\frac{q_2}{2k_2}x^2 + \frac{C_3}{k_2}x + C_4.$$